HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and built scalable backend infrastructure for abuse detection systems, integrating **AI/ML models** into production environments with a focus on **free-tier abuse prevention**.
- Architected **robust APIs** and **data pipelines** supporting **real-time and batch moderation workflows** for platform safety, processing approximately **1,500 moderation events per hour**.
- Implemented comprehensive **monitoring, alerting, and observability systems** using **Prometheus and Grafana**, establishing key **SLIs and SLOs** for safety infrastructure performance.
- Partnered with ML engineers to translate research models into production-ready systems, integrating them across the product suite and ensuring **scalable deployment**.
- Drove technical decisions and contributed vision to the safety roadmap, focusing on the next generation of platform guardrails for **scale and precision**.
- Took ownership of a new **fraud detection system** from 0→1, including defining requirements, building the core backend services, and deploying it to production, resulting in a **22% reduction in identified fraudulent accounts** within the first quarter.
- Containerized backend services using **Docker** and orchestrated deployments on **Kubernetes (EKS)**, managing CI/CD pipelines with **Jenkins** for automated testing and release cycles.

SKILLS

Backend Engineering – Python, Distributed Systems, APIs, Data Pipelines, Asynchronous Python, Backend Frameworks, Microservices, Java Spring Boot, REST APIs

AI & ML – AI Models, ML Models, MLOps, Fraud Detection, Model Deployment, Model Integration, Inference

Infrastructure & DevOps – AWS, GCP, Docker, Kubernetes, EKS, CI/CD, Jenkins, Prometheus, Grafana, Observability, SLIs, SLOs

Data & Analytics – SQL, Data Analysis, Real-time Streaming, Event-Driven Architectures

Safety & Trust – Abuse Detection, Free Tier Abuse, Moderation, Guardrail Systems, Content Moderation, Integrity Engineering

PROJECTS

Free-Tier Abuse Detection Platform

- Designed and implemented a scalable backend service using **Python (FastAPI)** to ingest and analyze user activity data, flagging suspicious patterns indicative of free-tier abuse.
- Integrated a **fraud detection ML model** into the pipeline, achieving **~0.91 AUC** on a held-out dataset of simulated free-tier abuse scenarios.
- Developed **real-time alerting mechanisms** and a dashboard for monitoring abuse trends, enabling proactive intervention and reducing automated account creation by an estimated **30%**.

Production Safety Guardrail Implementation

- Built RESTful APIs with **Java Spring Boot** to enforce safety policies across user-generated content, processing approximately **500 requests per second**.
- Implemented comprehensive **logging and monitoring** for the guardrail system using **Splunk**, ensuring high availability and rapid incident response.
- Collaborated with operations teams to define **SLOs** and integrate the system into existing production workflows, enhancing platform integrity.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)